

1. Descriptive Statistics & Foundations

1.1 Measures of Central Tendency & Dispersion

- Sample Mean ($\bar{x}$):

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Arithmetic average of all data points.

- Sample Variance (s^2) & Standard Deviation (s):

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}$$

s^2 measures mean squared deviation; s is in original data units.

1.2 Hypothesis Testing & Decision Error Framework

Decision Made	H0 is True (No Effect)	H0 is False (Effect Exists)
Reject H_0	Type I Error (α) — False Positive	Correct Decision (Power = $1 - \beta$)
Fail to Reject H_0	Correct Decision ($1 - \alpha$)	Type II Error (β) — False Negative

2. Inferential Hypothesis Tests & Parametric Checks

2.1 Shapiro–Wilk Test (Normality Check)

Evaluates whether sample data is consistent with a normal distribution.

Formula:

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)}\right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Symbol	Meaning	Role / Interpretation
n	Sample size	Total number of observations

Symbol	Meaning	Role / Interpretation
x_i	Original observation	Data value at index i
$x_{(i)}$	i -th order statistic	Data point after sorting in ascending order
$\bar{x}$	Sample mean	Arithmetic mean of sample
a_i	Shapiro–Wilk coefficient	Weights derived from normal distribution covariances
W	Test statistic	Range $[0,1]$. Values close to 1 indicate normality
p -value	Significance probability	Reject H_0 if $p < \alpha$ (commonly 0.05) $\rightarrow$ non-normal distribution

2.2 Levene's Test (Homogeneity of Variance)

Tests whether k independent groups possess equal population variances (homoscedasticity).

Formula:

- **Step 1 (Deviations):**

$$z_{ij} = |x_{ij} - \bar{x}_i|$$

- **Step 2 (F -Statistic):**

$$W = \frac{(N - k) \sum_{i=1}^k n_i (\bar{z}_i - \bar{z})^2}{(k - 1) \sum_{i=1}^k \sum_{j=1}^{n_i} (z_{ij} - \bar{z}_i)^2}$$

Under H_0 , W follows an F -distribution with $df_1 = k - 1$ and $df_2 = N - k$.

Symbol	Meaning	Role / Interpretation
k	Number of groups	Total independent comparison groups

Symbol	Meaning	Role / Interpretation
n_i	Group sample size	Number of observations in group i
N	Total sample size	Sum of all group observations ($N = \sum n_i$)
x_{ij}	Individual observation	j -th observation in group i
$\bar{x}_i$	Group center	Mean or median (Brown–Forsythe) of group i
z_{ij}	Absolute deviation	Deviation of observation from group center
$\bar{z}_i$	Group mean deviation	Average of deviations in group i
$\bar{z}$	Grand mean deviation	Overall average of all deviations across all groups
df_1, df_2	Degrees of freedom	$df_1 = k - 1, \quad df_2 = N - k$

2.3 Student's t -Tests (One-Sample & Welch's Two-Sample)

Parametric tests comparing group means.

Formulas:

- **One-Sample t -Test:**

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- **Welch's Two-Sample t -Test:**

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Welch's t -test does not assume equal variances between the two groups.

Symbol	Meaning	Role / Interpretation
$\bar{x}, \bar{x}_1, \bar{x}_2$	Sample means	Observed averages for sample, group 1, and group 2
μ_0	Hypothesized mean	Reference population baseline value
s, s_1, s_2	Sample standard deviations	Spread of sample, group 1, and group 2
n, n_1, n_2	Sample sizes	Number of observations in each sample
t	t -statistic	Distance from hypothesized value in standard error units
df	Degrees of freedom	One-sample: $df = n - 1$; Welch: Welch–Satterthwaite formula

2.4 Mann–Whitney U Test (Non-Parametric Group Comparison)

Non-parametric alternative to the independent two-sample t -test.

Formulas:

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1, \quad U_2 = n_1 n_2 + \frac{n_2(n_2 + 1)}{2} - R_2$$

$$U = \min(U_1, U_2)$$

Calculates rank sums after pooling and ordering data from both groups.

Symbol	Meaning	Role / Interpretation
n_1, n_2	Sample sizes	Number of observations in Group 1 and Group 2
R_1, R_2	Rank sums	Sum of ranks assigned to Group 1 and Group 2

Symbol	Meaning	Role / Interpretation
U_1, U_2	U statistics	Group-specific rank score differences
U	Final test statistic	Minimum of U_1 and U_2 used for critical value lookup

2.5 Chi-Square (χ^2) Test for Independence

Compares observed frequency counts against expected frequencies.

Formula:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Symbol	Meaning	Role / Interpretation
O_i	Observed frequency	Actual count in category/cell i
E_i	Expected frequency	Expected count under H_0 assumption
k	Number of categories	Total cells in contingency table
df	Degrees of freedom	$df = (\text{Rows} - 1) \times (\text{Columns} - 1)$

2.6 One-Way ANOVA (Comparison of 3+ Group Means)

Determines whether statistically significant mean differences exist across three or more independent groups.

Formulas:

$$MS_B = \frac{SS_B}{k - 1} \quad \text{where } SS_B = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2$$

$$MS_W = \frac{SS_W}{N - k} \quad \text{where } SS_W = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

$$F = \frac{MS_B}{MS_W}$$

Symbol	Meaning	Role / Interpretation
k	Number of groups	Total comparison group categories
N	Total observations	Combined sample size ($N = \sum n_i$)
$\bar{x}_i$	Group mean	Average value of group i
$\bar{x}$	Grand mean	Overall mean of all combined data
SS_B, SS_W	Sums of squares	Between-group (SS_B) and Within-group / Error (SS_W)
MS_B, MS_W	Mean squares	$SS_B/(k - 1)$ and $SS_W/(N - k)$
F	ANOVA test statistic	Follows $F(k - 1, N - k)$ distribution under H_0

3. Linear Regression & Diagnostic Checks

3.1 Model Goodness-of-Fit (R^2 and Adjusted R^2)

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- **Adjusted R^2 :**

$$\text{Adjusted } R^2 = 1 - \left[\frac{(1 - R^2)(N - 1)}{N - p - 1} \right]$$

R^2 measures variance explained; Adjusted R^2 penalizes adding non-informative predictors (p = predictor count).

3.2 Gauss–Markov Classical Assumptions & Diagnostics

Assumption	Core Mathematical Condition	Diagnostic Method	Remedial Strategy
1. Linearity	$E[\varepsilon_i X] = 0$ (Linear in parameters)	Residuals vs. Fitted plot (Look for random scatter)	Log, square-root, or polynomial feature transforms
2. Homoscedasticity	$\text{Var}(\varepsilon_i X) = \sigma^2$ (Constant error variance)	Levene's Test, Breusch–Pagan Test, Scale-Location plot	Use White/HC3 Robust Standard Errors or Weighted Least Squares (WLS)
3. Normality of Residuals	$\varepsilon_i \sim N(0, \sigma^2)$ (Errors normally distributed)	Shapiro–Wilk test on residuals, Q-Q Plot	Log/Box-Cox transform on response variable Y
4. No Multicollinearity	$\text{Corr}(X_i, X_j) < 1$ (No perfect linear dependence)	Variance Inflation Factor: $\text{VIF}_j = \frac{1}{1-R_j^2}$	Remove correlated predictor or use Ridge / PCA